Next Toppers

आरंभ BATCH

ACP

AARAMBH CHAPTERWISE PROBLEMS

Class-9th

Improvement in Food Resources

Improvement in Food Resources

(Options acche se padhna)

MULTIPLE CHOICE QUESTIONS (1 Mark)

1. Which one is NOT a source of carbohydrates?
(a) Rice
(b) Millets
(c) Sorghum
(d) Gram

2 . To solve the food problem of the country, which among the following is necessary?
(a) Increased production and storage of food grains
(b) Easy access of people to the food grain
(c) People should have money to purchase the grains
(d) All of the above

3.Find out the correct sentences.
(I) Hybridisation means crossing between genetically dissimilar plants.
(II) Cross between two varieties is called intraspecific hybridisation.
(III) Introducing genes of the desired character into a plant gives genetically modified crops.
(IV) Cross between plants of two species is called intervarietal hybridisation.
(a) (I) and (II)
(b) (I) and (IV)
(c) (I) and (III)
(d) (II) and (IV)

4. Which of the following is called the surface feeder?
(a) Catla
(b) Rohu
(c) Mrigal
(d) Grass Carp

5. Cyperus and Parthenium are types of:
(a) Diseases
(b) Pesticides
(c) Weeds
(d) Pathogens

6. Crops which are grown in the month of June to October are called:
(a) Kharif
(b) Rabi
(c) Zaid
(d) Both (a) and (b)

7. Weeds affect the crop plants because they grow:
(a) Killing of plants in field
(b) Dominating the plants to grow
(c) Competing for various resources of crops (plants) causing low availability of nutrients
(d) All of the above

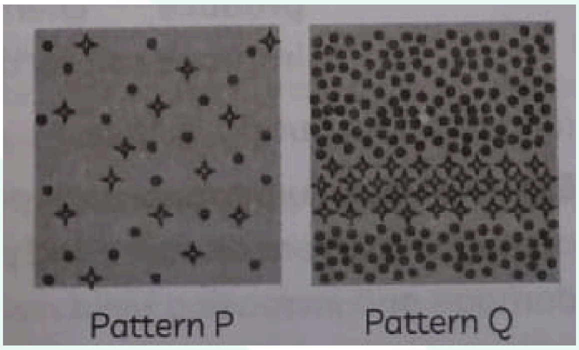
8. What are the desirable agronomic characteristics for crop improvements?
(a) Dwarfness is desired in cereals
(b) Tallness and profuse branching for fodder crops
(c) Both (a) and (b)
(d) None of the above

9. Find out the correct sentence about manure.
(I) Manure contains large quantities of organic matter and small quantities of nutrients.
(II) It increases the water holding capacity of sandy soil.
(III) It helps in draining out of excess water from clayey soil.
(IV) Its excessive use pollutes the environment because it is made of animal excretory waste
(a) (I) and (II)
(b) (I) and (III)
(c) (II) and (III)
(d) (III) and (IV)

10. Animal husbandry is the scientific management of:
(I) Animal breeding
(II) Culture of animals

(III) livestock
(IV) Rearing of animals
(a) (I), (II) and (III)
(b) (II), (III) and (IV)
(c) (I), (II) and (IV)
(d) (I), (III) and (IV)

11.. Illustrates two distinct cropping patterns, P and Q. Identify these cropping methods and determine which statement about them is INCORRECT.



(a) In cropping pattern P, crops should have different root patterns.
(b) Crops chosen for cropping pattern Q should require different types of nutrients.
(c) In cropping pattern P, mustard and wheat can be grown together.
(d) In cropping pattern Q, legumes such as cowpea, mung bean, and soybean should be cultivated simultaneously.

(yaha Marks katate h)

Assertion And Reasons Questions

These consist of two statements - Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:
(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

12. Assertion (A): Fungicides and pesticides increase crop output.
Reason (R): Manure and fertilizers produce chemicals that improve soil fertility.

13. Assertion (A): Composite fish culture helps in maximising fish production.
Reason (R): In composite fish culture, different species of fish are selected based on their feeding habits to avoid competition for food.

14. Assertion (A): The management of poultry farms requires light.
Reason (R): For eggs to be produced at their maximum, 14–16 hours of light per day is necessary.

15. Assertion (A): A. mellifera bee is used for commercial honey production.
Reason (R): A. mellifera has a high honey collection capacity and also they breed very well.

16. Assertion (A): Layers receive more vitamins K and A.
Reason (R): To produce eggs, layers are raised.

(pahle points socho firr likho)

SHORT ANSWER TYPE QUESTIONS (2 - 3 Marks)

17. A farmer wants to improve the cattle breed for getting a higher yield of milk and resistance to diseases.
Name the method he should adopt to get animals of desired qualities. Illustrate it for the cows.

18. List any two factors for which crop variety improvement is done.

19. (A) Why is excessive use of pesticides not advisable?
(B) Define vermi-composting.

20. (A) In what ways do honeybees aid in cross-fertilisation?
(B) Which is the most advantageous fish culture system?

21. Mohan told his friend Deepak, who runs a poultry farm, that broiler production is indeed a solution to increase the production of nutritious animal protein food.
Support Mohan's view by listing any four factors that need to be considered for broiler production.

22. Differentiate between compost and vermi-compost.

23. What is a GM crop? Name any one GM crop which is grown in India.

24. If there is low rainfall in villages throughout the year, what measures will you suggest to the farmers for better cropping?

25. In terms of cattle feed, how do roughage and concentrates differ from one another?

26. (A) Why should preventive methods and biological control methods be preferred for protecting crops?
(B) Name the farming system in which only such above-mentioned methods are followed.
(C) How does manure improve the soil structure of sandy and clayey soil?

27. What would happen if poultry birds are larger in size and do not have summer adaptation capacity? In order to get small-sized poultry birds having summer adaptability, what methods will be employed?

28. What is the difference between chemical fertilizers and composts?

29. Ravi saw that every year his grandfather used leguminous crops before sowing new crops.
His grandfather replied it is desirable in crop rotation.
What's the reason behind it?

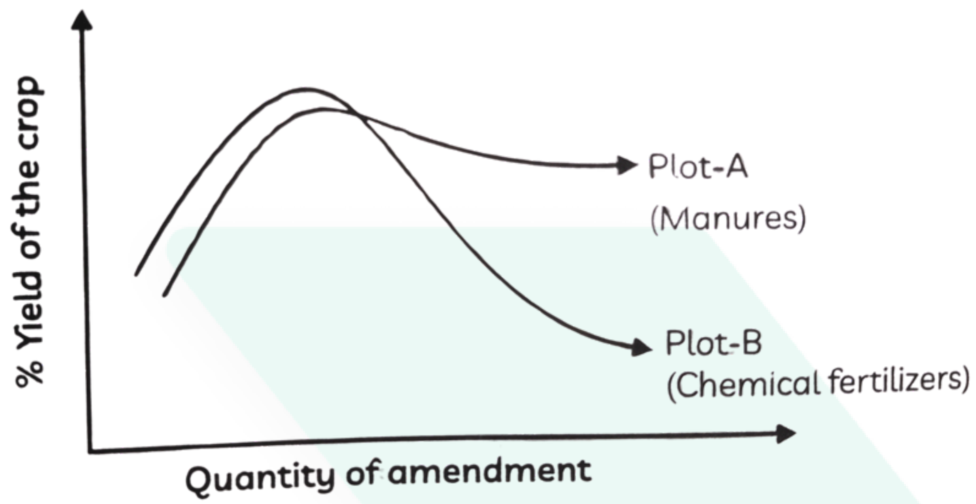
30. What are common techniques in beekeeping, fishing, and poultry for boosting production?

(pahle points socho firr likho)

Long Answer Type Questions (5 Marks)

31. The figure shows two crop fields [Plots A and B] that have been treated with manures and chemical fertilizers respectively, keeping other environmental factors the same. Observe the graph and answer the following questions:
(A) Why does Plot B show a sudden increase and then a gradual decrease in yield?
(B) Why does Plot A show a steady, slightly delayed increase?

(C) What is the reason for the different patterns of the two graphs?



32. Why is crop variety improvement important for cultivation?
Describe the important factors for which crop variety improvement is done.

33. Ramu read in the newspaper: "Storage of grains in India is still a big problem." She asked her teacher to explain how storage of grains is related to food security. Explain the reasons why stored food grains sometimes get damaged. Suggest preventive and control measures.

34. Differentiate between manure and fertilizer.

35. Differentiate between capture fishery and culture fishery. (Any five points)

(read paragraph first)

Case Study/Source Based Question

36. A farmer in Madhya Pradesh grows wheat and mustard in the same field. He practices this pattern every year. Recently, he read about other methods like intercropping and crop rotation in a government training program. He observed that his yield is not increasing much and that pests are also affecting the crops frequently.
He is now considering using organic manure instead of relying only on chemical fertilizers.

Based on this case, answer the following questions:

(A) Which cropping pattern is the farmer following?
(B) Differentiate between intercropping and crop rotation.
(C) Why are preventive and biological methods of crop protection better than chemical methods?
(D) How does the use of manure improve soil structure compared to fertilizers?

37. Ravi's uncle runs a poultry farm and his neighbor manages a fish culture pond. Ravi noticed that poultry birds in his uncle's farm often suffer from heat stress during summer. His uncle has to spend extra money on cooling arrangements. On the other hand, his neighbor uses composite fish culture in which different species of fish such as Catla, Rohu, and Mrigal are reared together. The neighbor earns more profit with fewer losses.

Based on this case, answer the following questions:

(A) Why do poultry birds with larger body size have less adaptability to summer?
(B) Mention two factors that are considered for successful broiler production.
(C) What is composite fish culture? How does it give better yield?
(D) Give one advantage and one disadvantage of using chemical fertilizers in fish ponds.
(E) Suggest two measures Ravi's uncle can adopt to maintain egg and meat production during summers.

Chapter Special [KBC]

38. A farmer used chemical fertilizers continuously for 5 years. When he switched back to only manure, his soil became softer, but the crop yield decreased.
(a) Explain why the yield fell despite improved soil structure.
(b) Suggest how the farmer can restore high yield sustainably.

39. A fish farmer stocked Catla, Rohu, and Silver Carp together. Soon, competition for food increased and yield fell.
(a) Identify the scientific mistake.
(b) State the correct principle of composite fish culture he should follow.

40. Two farmers grew wheat and mustard in alternate rows. One had fewer pests, while the other had heavy infestation.
(a) Give two scientific reasons why their outcomes differed.
(b) Suggest one precaution for successful intercropping.

41. Farmers using pesticides noticed that after a few years, pest populations increased instead of decreasing.
(a) Explain scientifically why this paradox occurs.
(b) Suggest an alternative method of pest control.

42. A farmer sowed leguminous crops expecting nitrogen fixation, but soil nitrogen remained low.
(a) Suggest two reasons why nitrogen fixation might have failed.
(b) Name the bacteria responsible for nitrogen fixation in legumes.

43. A farmer reported fungal infection in seedlings after using vermicompost, even though it usually improves soil quality.
(a) Explain scientifically how this could happen.
(b) Suggest two precautions to avoid such problems.

SOLUTIONS

MULTIPLE CHOICE TYPE QUESTIONS

1. D
2. D
3. C
4. A
5. C
6. A
7. C
8. C
9. A
10. D
11. D

ASSERTION AND REASONS TYPE QUESTIONS:

12. b
13. a
14. a
15. a
16. d

SHORT ANSWER TYPE QUESTIONS

17. The farmer should adopt the method of cross-breeding.

- **Cross-breeding means mating indigenous (local) breeds of cattle with exotic (foreign) breeds to combine desirable qualities of both.**
- **Example in cows:**
  - **An Indian breed such as Sahiwal (good disease resistance and adaptability to local climate) can be crossed with an exotic breed such as Holstein-Friesian (high milk yield).**
  - **The resulting offspring will have improved milk production along with better resistance to diseases.**

Thus, by cross-breeding, the farmer can obtain cattle with both high milk yield and disease resistance.

18. Crop variety improvement is done mainly for the following factors:

1. **Higher yield – to increase the productivity of crops per unit area.**
2. **Resistance to diseases and pests – to reduce crop losses and dependence on pesticides.**

SOLUTIONS

19. (A) Excessive use of pesticides is not advisable because:

1. **It causes soil and water pollution.**
2. **Pesticides kill beneficial organisms along with harmful ones.**
3. **Overuse may make pests resistant, reducing effectiveness.**
4. **Harmful pesticide residues may enter the food chain and affect human health.**

(B) Vermi-composting is the process of preparing compost using earthworms.

- **Earthworms help decompose organic matter (like farm waste, dung, and kitchen waste) into nutrient-rich manure called vermi-compost, which improves soil fertility.**

20. (A) Honeybees act as pollinating agents. While collecting nectar and pollen from flowers:

1. **They transfer pollen grains from the anther of one flower to the stigma of another flower.**
2. **This helps in cross-pollination, which leads to cross-fertilisation in plants.**
3. **Cross-fertilisation produces genetically better seeds and fruits with desirable qualities.**

(B) The most advantageous fish culture system is Composite Fish Culture.

- **In this system, different species of fish (e.g., Catla, Rohu, Mrigal, Grass Carp, Silver Carp) are cultured together in the same pond.**
- **These fishes have different feeding habits (surface, middle, bottom feeders), so there is no competition for food.**
- **This results in better utilization of all food resources in the pond and a higher yield of fish.**

21. Mohan is correct because broiler production provides a good source of nutritious animal protein food. The following factors need to be considered for successful broiler production:

1. **Nutrient-rich diet – Broilers should be given a diet rich in proteins and adequate fat to promote rapid growth.**
2. **Proper housing – Well-ventilated, hygienic, and spacious housing is required to reduce stress and prevent diseases.**
3. **Health care – Broilers need regular vaccination and protection from common poultry diseases.**
4. **Suitable environment – Broilers should be raised in conditions with proper temperature, light, and cleanliness to ensure healthy growth.**

SOLUTIONS

22.

Compost	Vermi-compost
Compost is prepared by the decomposition of organic matter (plant waste, animal waste, kitchen waste) by microorganisms.	Vermi-compost is prepared by the decomposition of organic matter with the help of earthworms.
It takes a longer time to form.	It forms faster because earthworms speed up decomposition.
Nutrient content is moderate.	Nutrient content is richer and more easily available to plants.
Quality may vary depending on raw material and microbes.	Quality is generally uniform and superior because of earthworm activity.

23. A GM crop (Genetically Modified crop) is a crop in which the genes are altered using biotechnology to introduce desirable traits such as pest resistance, higher yield, or tolerance to environmental stress.

- **Example in India: Bt cotton – It has been genetically modified to resist bollworm pests.**

24.In case of low rainfall, farmers can adopt the following measures for better cropping:

1. **Use drought-resistant varieties – Grow crops that can survive with less water (e.g., millets, pulses).**
2. **Efficient irrigation methods – Adopt water-saving methods like drip irrigation and sprinkler irrigation.**
3. **Rainwater harvesting – Collect and store rainwater in ponds or tanks for later use.**
4. **Soil moisture conservation – Use mulching and proper tillage practices to retain soil moisture.**
5. **Crop rotation and intercropping – Grow crops that improve soil fertility and require less water.**

आरंभ BATCH

SOLUTIONS

25. Roughage is high in fibre and low in nutrients. It mainly consists of bulky plant materials like hay, straw, and green fodder. Roughage aids digestion, supports rumen function, and provides the necessary bulk in the diet.

- **Concentrates, on the other hand, are low in fibre but rich in carbohydrates, proteins, vitamins, and minerals. Examples include grains like oats and barley, and by–products such as wheat bran and molasses. Concentrates supply concentrated energy and nutrients that are essential for growth, milk production, and fattening.**

26. (A) They are eco-friendly, avoid pollution, maintain ecological balance, and are safer than chemical pesticides.
(B) Organic farming.
(C) In sandy soil, it increases water-holding capacity.
In clayey soil, it makes the soil loose, porous, and well aerated.

27. Larger, non-adapted birds will suffer from heat stress, reduced egg laying, poor growth, low fertility, and higher mortality in summers.
Cross-breeding and selective breeding methods are used to develop small-sized poultry with better summer adaptability and higher productivity.

28.

Point	Chemical Fertilizers	Composts
Nature	Inorganic, factory-made	Organic, formed by decomposition of waste
Nutrients	Supply few nutrients (mainly N, P, K) quickly	Supply all nutrients slowly and evenly
Effect on Soil	May reduce soil fertility on long use	Improve soil texture and fertility
Environment Impact	Can cause pollution of soil and water	Eco-friendly, no pollution

SOLUTIONS

29. Leguminous crops (like peas, gram, beans) have root nodules containing nitrogen-fixing bacteria (Rhizobium).
→ These bacteria convert atmospheric nitrogen into usable nitrogen compounds.
→ This naturally increases soil fertility, reducing the need for chemical fertilizers for the next crop.

30. Beekeeping → Use of improved breeds of bees, proper hive management, and prevention of diseases.
Fishing → Composite fish culture, aquaculture, use of high-yielding fish varieties.
Poultry → Cross-breeding for better egg/meat production, disease control, and improved housing and feed.

31. (A) Plot B was treated with chemical fertilizers. They supply nutrients quickly, so crop yield rises suddenly. But long-term use reduces soil fertility and kills useful microbes, leading to a gradual decrease in yield.
(B) Plot A was treated with manure. Manures release nutrients slowly, improve soil texture, water-holding capacity, and increase soil fertility gradually. Hence, yield increases steadily, though with some delay.
(C) Reason for different patterns:

- Chemical fertilizers → Quick but short-term effect, harmful in the long run.
- Manures → Slow but long-lasting improvement in soil fertility and sustainable yield

32. Crop variety improvement is important to increase crop yield, improve quality, make crops resistant to diseases and pests, adapt them to environmental stresses, and reduce dependence on chemical inputs.
<u>Important factors for which crop variety improvement is done:</u>

1. Higher yield – To produce more food per unit area.
2. Improved quality – Better taste, nutrition (proteins, oils, vitamins), and storage.
3. Biotic resistance – Resistance to diseases, insects, and pests.
4. Abiotic resistance – Tolerance to drought, salinity, flood, and extreme temperatures.
5. Shorter duration / early maturity – So that multiple crops can be grown in a year.
6. Wider adaptability – So the crop can be grown in different climatic and soil conditions.
7. Desirable agronomic traits – Such as tallness in fodder crops, dwarfness in cereals, or more tillering in wheat.

SOLUTIONS

33. Proper storage of food grains is essential for food security because grains are grown seasonally but consumed throughout the year. If grains get damaged or spoiled during storage, there will be shortage of food supply and loss to farmers and consumers.

Reasons why stored food grains sometimes get damaged:

1. **Biotic factors – Insects, rodents, birds, bacteria, and fungi.**
2. **Abiotic factors – Moisture, temperature, and poor storage conditions.**
3. **Improper drying before storage.**
4. **Lack of scientific storage facilities.**

Preventive and control measures:

1. **Proper drying of grains before storage.**
2. **Cleaning and regular inspection of godowns and silos.**
3. **Use of insecticides and fumigation in storage areas.**
4. **Maintaining low moisture and proper aeration.**
5. **Use of modern storage structures like metal bins, silos, and warehouses.**
6. **Encouraging safe traditional methods (like neem leaves, ash) in rural areas.**

34.

Point	Manure	Fertilizer
Nature	Natural, organic substance from decomposition of plant/animal waste	Manufactured in factories; may be inorganic (chemical) or organic
Nutrient Content	Low in nutrients but contains all essential elements in small amounts	Rich in specific nutrients like N, P, K
Effect on Soil	Improves soil structure, water-holding capacity, and long-term fertility	Does not improve soil texture; long use may reduce fertility
Action	Nutrients released slowly, effective for a longer duration	Nutrients available immediately, but effect is short-lived
Environment	Eco-friendly, no pollution	Excess use causes soil and water pollution

SOLUTIONS

35.

Point	Capture Fishery	Culture Fishery
Definition	Fishing from natural water bodies like rivers, seas, oceans	Rearing and breeding of fish in man-made or controlled water bodies
Source	Depends on natural stocks of fish	Fish are raised in ponds, tanks, or reservoirs
Control	Uncontrolled, depends on availability in nature	Controlled, as fish are managed and cultivated
Production	Production is uncertain and limited	Production is high and predictable
Examples	Marine fishing, river fishing	Aquaculture, composite fish culture

CASE STUDY/SOURCE BASED QUESTION:

36. (A) Mixed cropping.
(B) Intercropping → Growing two or more crops simultaneously in the same field in a definite row pattern (e.g., maize + soybean).
Crop Rotation → Growing different crops one after another in the same field in different seasons (e.g., rice → wheat → pulses).
(C) They are eco-friendly, cost-effective, do not pollute soil and water, do not kill useful organisms, and maintain ecological balance.
(D) Manure improves soil texture, increases water-holding capacity in sandy soil, makes clayey soil loose and porous, and enhances long-term fertility. Fertilizers give quick nutrients but may damage soil health in the long run.

SOLUTIONS

37. (A) Poultry birds with larger body size have less surface area relative to body volume, so they cannot lose body heat easily and show poor adaptability to summer.
(B) Two factors for successful broiler production:
Proper feed with high protein and vitamin content.
Suitable housing and disease control.
(C) Composite fish culture is the rearing of different species of fish (e.g., Catla, Rohu, Mrigal) together in the same pond. Since these fishes occupy different food zones, there is no competition, and maximum use of food resources gives better yield.
(D) Advantage: Fertilizers promote plankton growth, providing natural food for fish.
Disadvantage: Excess use causes water pollution and oxygen depletion in ponds.
(E) Two measures for summer:
Provide summer-adapted, small-sized breeds of poultry.
Improve housing by proper ventilation, cooling, and clean drinking water.

PK Special [KBC]

38. (a) Yield fell because manure contains nutrients in small amounts and releases them slowly. After using chemical fertilizers for 5 years, the soil lacked sufficient readily available nutrients, so crops could not get enough nutrition even though the soil structure improved.
(b) The farmer can restore high yield sustainably by using a combination of manures and chemical fertilizers, along with crop rotation (especially legumes), intercropping, and organic practices to maintain both soil fertility and productivity.

39. (a) The mistake was that Catla, Rohu, and Silver Carp all feed in the upper water layer, so they competed for the same food.
(b) In composite fish culture, species should be selected that occupy different feeding zones (surface, middle, and bottom) of the pond — e.g., Catla (surface feeder), Rohu (middle feeder), and Mrigal or Common Carp (bottom feeder) — so that food resources are used fully without competition.

40.(a) In intercropping, alternate rows of different crops reduce the spread of pests and diseases, as pests cannot easily move from one row to another. Different crops utilize nutrients differently, preventing excess growth of pests that thrive on a single crop.
(b) For successful intercropping, crops chosen should have different nutrient requirements and growth habits (e.g., deep-rooted + shallow-rooted).

SOLUTIONS

41. (a) Over time, some pests develop resistance to pesticides due to repeated exposure. The natural enemies (predators/parasites) of pests are also killed by pesticides. This allows resistant pests to multiply rapidly, so their population increases instead of decreasing.
(b) An alternative method is biological control — using natural predators, parasites, or microorganisms to control pest populations (e.g., ladybird beetles to control aphids).

42. (a) Nitrogen fixation might have failed because:
1. **Root nodules were not formed due to absence of proper Rhizobium strain in the soil.**
2. **Poor soil conditions (like low aeration, improper pH, or waterlogging) prevented the growth of nitrogen-fixing bacteria.**

(b) The bacteria responsible are Rhizobium.

43. (a) Fungal infection could happen if the vermicompost was not properly prepared or stored. Incomplete decomposition or excess moisture may allow harmful fungi to survive and infect seedlings, even though vermicompost normally improves soil health.
(b) Precautions:
1. **Ensure proper decomposition and curing of vermicompost before use.**
2. **Keep vermicompost dry, well-aerated, and free from contamination during preparation and storage.**